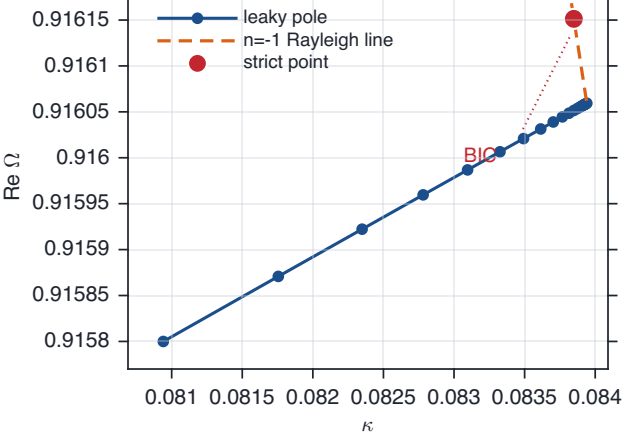
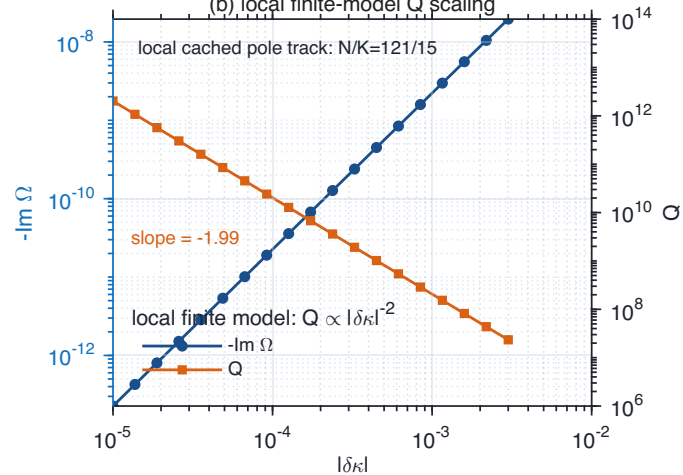


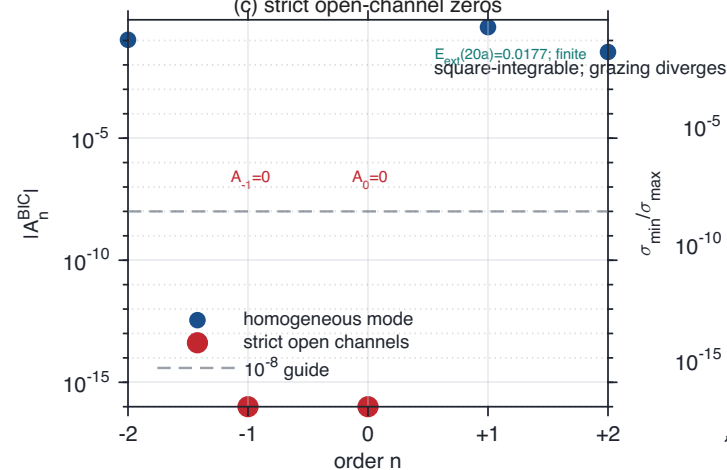
(a) pole meets the Rayleigh branch point



(b) local finite-model Q scaling



(c) strict open-channel zeros



(d) two-groove root convergence

